

Progress Report Sprint 8

- Course Name: CS461 Senior Software Engineering Project I
- Date: 02/12/26
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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Summary - Outcomes, blockers, and insights

- Completed two UI-related bug fixes, styled the tutorial pop-ups, and added interaction from the application behaviors with the different phases of the tutorial
- Created framework for second taxonomy pass using NCBI API.
- Made substantial improvements to the performance of the Decision Tree Regressor.
- We met our goal of having algorithm-related code pushed to the GitHub repository by the end of this sprint, so we are satisfied with our progress on that front

Blockers:

- Faced issues with tutorial implementation due to the 'Interconnected JS Movements' bug and the 'Mismatched Element IDs' bug; overcame this by patching both bugs to work more effectively on the tutorial feature.
- We need further parsing and API integration to fully complete the NCBI API for the second taxonomy dataset pass.

Work Log

1. Tutorial Styling and Connected UI Behaviors

Status: Complete

Owner: Hailey

Description: Last sprint, I implemented the tutorial skeleton, completing the content of each tutorial pop-up and the basic javascript. This sprint, I styled the pop-ups, decided placements for them, and added in movements for each element that matched the current tutorial information.

Source Link: [w 2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifact: Item.01

Notes: Although it may be tweaked slightly as features of the UI change, the implementation for the website as it is now is considered 'complete'.

2. UI Bug Fixes

Status: Complete

Owner: Hailey

Description: Last sprint, I owned three open bugs related to our project's UI, so I resolved to fix two of them. I, firstly, fixed the 'Mismatched Element IDs' bug; this issue was caused by the skeleton code for the tutorial being written before the content, so when the content was plugged in, a lot of the IDs didn't make sense. Changing each ID also meant altering the javascript movements to accommodate the changes. Next, I patched the 'Interconnected JS Movements' bug; this bug was focused around the click event handlers in index.js. Sometimes, when an element becoming visible relied on a different element being closed, there would be user scenarios in which the element wouldn't show up at all. I went through and made the movements less reliant on other element statuses.

Source Link:  TaxonBodyMassML Risk Tracker

Supporting Artifact: Item.02 and Item.03

3. Decision Tree Regression

Status: In-progress

Owner: Grant

Description: The decision tree now operates in the log space. This avoids the previous issue of truncation of small mass species during prediction. Furthermore, we improved the performance of the model on the test set by fine tuning some hyperparameters.

Source Link: [2026-01-13-CS.032-Sprint6-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the XGBoost library, check out the manpages here: [XGBoost](#)

4. Taxonomy Data 2nd Pass

Status: In-progress

Owner: Grant

Description: The code for the 2nd dataset pass is almost complete, we just need to finish out the API connection with NCBI. For additional context, this program is responsible for finding any missing taxonomy data and filling it in with likely candidates from the NCBI API.

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the API, check out the manpages here: [NCBI API](#)

5. API Testing and Calling

Status: In-progress

Owner: Anish

Description: The code for the API is almost complete, we just to pass some instructions into the API and see if the json objects are being generated from the data set

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts: Item.04

Notes: For information about the API, check out the manpages here: [NCBI API](#)

Risk and Quality Tracking

Open Bug Count: 3

TOP BUGS	SEVERITY	OWNER(S)	STATUS	LINK	ACTIONS
DATASET TOO LARGE FOR DECISION TREE	High	Grant	In-progress	https://github.com/marknovak/TaxonBodyMassML/tree/random-forest/predictive_models/decision_tree_test.py	Since the model worked with about 30% of the dataset, as soon as the rest of the data is fixed, Grant will try running the model on a larger portion of the data.
NON-RESPONSIVE CSS	Low	Hailey	In-progress	primary CSS file on GitHub	Hailey will add in CSS that better accommodates smaller screen sizes (media queries) by the end of this term (not a top priority)
MISSING CLASS/ORDER FROM MANY DATASET ROWS	High	Grant	In-progress	https://github.com/marknovak/TaxonBodyMassML/tree/data-combination/data-combination	The API framework is set up.

Next Sprint Plan

- Focus on the 'Help' page of the web application. Clean up and complete functionality of the 'More Questions?' section, add content to the 'FAQs' section, and generally style and add more content to the whole page to aid in usability and to accommodate users with differing levels of technical literacy.
Priority: Medium; **Link:** [2025-10-23-CS.032-Requirements](#) (REQ-004)

- After using the NCBI API to fill in remaining missing values in the BodyMass dataset, we will run the existing model on the larger dataset. Then, we will also focus on tweaking the model to achieve higher accuracy. This will help us achieve the accuracy requirement:
Priority: High; Link: [2025-10-23-CS.032-Requirements\(REQ-008\)](#)

Team Reflection

This sprint, there were several substantial updates to the project. As for the UI, the tutorial feature is in a state we consider “complete”; this is a great step toward our requirements of accessibility and usability. We also patched two bugs from our risk tracker.

We were able to make substantial progress on the algorithm front. This mostly includes careful tuning of hyperparameters in the Decision Tree Regressor. We also made an architecture change by having the model predict values in the log scale. Furthermore, the first version of the LLM predictive model is complete. For now, it simply passes the taxonomy dataset to make predictions, but future versions will include phylogenetic data as well.

For the next sprint, we have all agreed that we’d like to see significant improvement on the ‘Help’ page. We should continue to work on our algorithm at the same time, but we know that, if we want a complete application to give to Mark at the end of next term, a comprehensive help page is necessary.

Artifacts Appendix

Item.01

[Video of Updated Tutorial](#)

This is a screen recording of the updated tutorial feature. The styling, as well as the associated movements for each tool-tip, is depicted in this video. At this point, this is considered a ‘complete’ implementation.

Item.02

```
<div id = "tutorial-start">
  <button id = "begin-tutorial-button" class = "hidden">Tutorial</button>
</div>
<div id = "input-bar-tutorial" class = "tool-tip hidden">
  <p>First, we'll need to know if you are inputting a single species, or
  <button id = "input-bar-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "single-tutorial" class = "tool-tip hidden">
  <p>If you are inputting a list of species names, please instead s
  <button id = "single-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "list-tutorial" class = "tool-tip hidden">
  <p>After clicking the 'List' button, you have an option to convert
  <button id = "list-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "csv-check-tutorial" class = "tool-tip hidden">
  <p>Now, you may go ahead and enter the name or list of names (each
  <button id = "csv-check-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
```

```
<div id = "tutorial-start">
  <button id = "begin-tutorial-button" class = "hidden">Tutorial</button>
</div>
<div id = "single-tutorial" class = "tool-tip hidden">
  <p>First, you will select whether you are inputting a single species, or
  <button class = "next-button" id = "single-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "list-tutorial" class = "tool-tip hidden">
  <p>If you are inputting a list of species names, you will instead s
  <button class = "next-button" id = "list-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "csv-check-tutorial" class = "tool-tip hidden">
  <p>After clicking the 'List' button, you have an option to convert
  <button class = "next-button" id = "csv-check-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
</div>
<div id = "input-bar-tutorial" class = "tool-tip hidden">
  <p>At this point, you will go ahead and enter the name or list of n
  <button class = "next-button" id = "input-bar-next">Next</button>
  <button class = "cancel">Exit Tutorial</button>
```

Fig.01 - Screenshot of bugged code; the “single-tutorial” id described the tutorial pop-up that tells users how to use the “list” option on the UI instead of the “single” option. This issue held for three other ids.

Fig.02 - Screenshot of fixed code; now, the “list-tutorial” id describes the tutorial pop-up that tells users how to use the “list” option on the UI, etc.

Item.03

Example of Bug on UI; Bug Fix

These are two recordings featuring 1) an example of the ‘Interconnected JS Movements’ bug and 2) that same example after the bug was fixed. You can see that, in the first video, because the intro modal was never actually closed by the user, the tutorial button never pops up; after the bug fix, the tutorial button is no longer reliant on the intro modal being closed, so it still shows up after clicking the ‘Home’ tab on the navigation bar.

Item.04

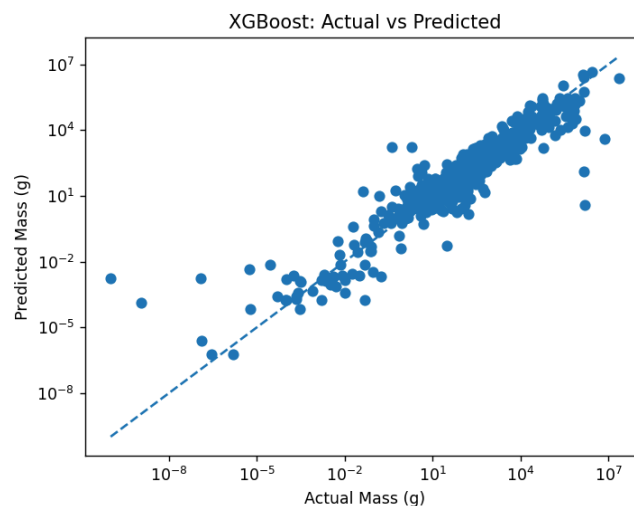


Fig.03 - This is a log-scale plot of the mass predictions for species in the test dataset. Since the last iterations of the plot, the accuracy on low-mass species has substantially improved due to running the model in $\log_{10}$. Furthermore, the accuracy has been improved via the tuning of hyperparameters.